MARK SCHEME · TEACHER COPY

11BR · Topic 2: Communication

Cambridge AS Level Computer Science 9618

Paper 1 Theory Fundamentals

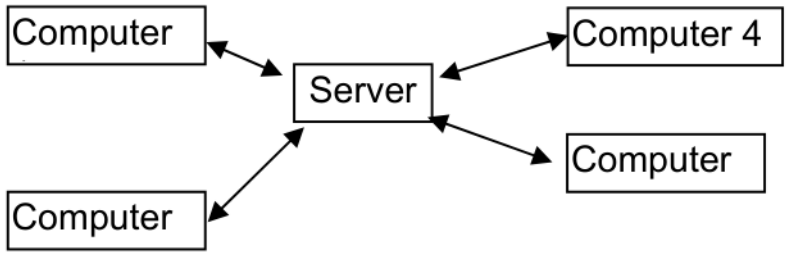
Total marks: 75

Official Cambridge mark scheme extracts for each question in the matching exam paper, with question numbers changed to match the exam. Source of each question is shown above its rows.

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Question 1 · May/June 2023 Paper 12, original Q1 (a),(b),(c),(d),(e) [11 marks]

Question	Answer	Marks
1(a)	**1 mark** each to **max 2** • Allows the sharing of files/data // Allows communication between the devices • Allows the sharing of resources e.g. hardware / software (applications) • Allows central management // by example, backup, security, etc.	**2**
1(b)	**1 mark** each to **max 2** • Covers a small geographical area • The infrastructure is privately owned // not controlled by external organisations	**2**
1(c)	**1 mark** each to **max 2** • Each computer directly connected only to the server ... • ... all components correctly labelled 	**2**

1(d)	**1 mark** each to **max 3** • A protocol (suite) • For **data transmission** over standard / universal **wired** / **cabled** network connections • Uses Carrier Sense Multiple Access / Collision Detection (CSMA/CD) • Data is transmitted in frames • … each frame has a source and destination (IP/MAC) address • ... and error checking data (so damaged frames can be resent)	**3**
1(e)	**1 mark** each • The server performs **minimal** / **some processing** for the client • The clients also do **most** of their own processing/work independently // most of the resources are installed locally	**2**

Question 2 · May/June 2023 Paper 13, original Q2 (a),(b),(c),(d),(e) [15 marks]

Question	Answer	Marks

Total: 75 marks

2(a)	**1 mark** each to **max 2** • WAN covers a large **geographical** area and LAN covers a small **geographical** area • LAN connections between devices are usually physical, whereas the WAN connections are often virtual • A LAN has a high data transfer rate, whereas a WAN has a low data transfer rate • The **ownership** of a LAN is private; the **ownership** of a WAN can be private or public • LAN is **usually** more secure than a WAN because protection is easier to implement	**2**
2(b)(i)	**1 mark** each to **max 2** • All computers are connected to at least one other device • There are multiple routes between devices • The computers can act as relays, passing packets on towards the final destination	**2**
2(b)(ii)	**1 mark** each to **max 2** • If one line goes down there are more routes available • Improved security as not using one main line • No/fewer collisions • New nodes can be added without interruption or interfering with other nodes • More secure because data is sent over a dedicated connection	**2**

2(c)	**1 mark** each to **max 2** • Server performs **all** processes required by the task and/or data storage • Clients **only** sends requests to the server and displays the returned results	**2**
2(d)	**1 mark** each to **max 4** Examples: • Some students might only have one sort of connection on their device • Wired provides better performance for the student's device • ... for example, enabling faster access to university databases • There will be less interference if students connect via a cable • Students can transmit private/confidential data/work securely • ... for example, their final dissertation • Wireless connection means that the students can use their devices in different rooms/sites/outside/anywhere more freely // student devices can be portable • Wireless connection enables the students to bring multiple devices // bring their own devices // change devices	**4**

2(e)	**1 mark** for identification • IP address is made up of a network ID and a host ID **1 mark** each to **max 2** for description • Each device in a subnetwork has the same network ID // Each subnetwork has a different network ID • Every device **in each subnetwork** has a different host ID but the same network ID // the host ID uniquely identifies the device **within the subnetwork**	**3**

Question 3 · May/June 2024 Paper 13, original Q5 (a),(b),(c),(d) [11 marks]

Question	Answer	Marks
3(a)(i)	**1 mark** for: Dedicated/bespoke services/storage on a remote server only available to company	**1**
3(a)(ii)	**1 mark** each to **max 3**: e.g. • Not reliant on a third party • … gives greater **control** over security/privacy • … gives greater **control** over backup • Storage can be tailored/scalable to company requirements// an example e.g. the amount of storage accessible/ facilitating the sharing of files	**3**

3(b)	**1 mark** each: • Sending computer transmits packets directly to switch/router/central device • Switch/router/central device **checks destination address** of packet and forwards directly to that device	**2**
3(c)(i)	**1 mark** each to **max 2**: • Jamming signal is transmitted by the **sending device** • Transmission is aborted • The **sending device** waits a **random** time before trying to send data again ... • ... if further collisions occur the wait time is increased	**2**
3(c)(ii)	**1 mark** each to **max 2**: • Random time increased each time so can be infinite waiting • May be constant jamming signal so nothing ever sends • Certain nodes cannot be prioritised • High power consumption • Only suitable for short distance network // limited distance • Not scalable // more nodes means exponentially longer waiting times	**2**
3(d)	**1 mark** for: Static means the IP for that device does not change **and** Private means it can only be accessed/seen/used within the LAN	**1**

Question 4 · May/June 2024 Paper 13, original Q7 (f) [6 marks]

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>4(a)(i)</td><td>1 mark for each correct method and
1 mark for corresponding description to max 4:

<table><tr><th>Method</th><th>Description</th></tr><tr><td>Parity byte</td><td>An additional bit is added to make the number of 1s in the byte odd or even to match the parity. If a byte with an odd number of 1 bits is received when even parity is used, there is an error.</td></tr><tr><td>Parity block</td><td>Parity is calculated horizontally and vertically. A parity byte is created from the bits produced by the vertical parity check. This is sent with the data. The parity is re-checked when received and the position of an incorrect bit can be determined.</td></tr><tr><td>Checksum</td><td>A calculation is made from the data and the result transmitted with the data. The receiver repeats the calculation and compares the result with the value received. If the two are different, there is an error.</td></tr></table></td><td>4</td></tr>
<tr><td>4(a)(ii)</td><td>1 mark each to max 2:

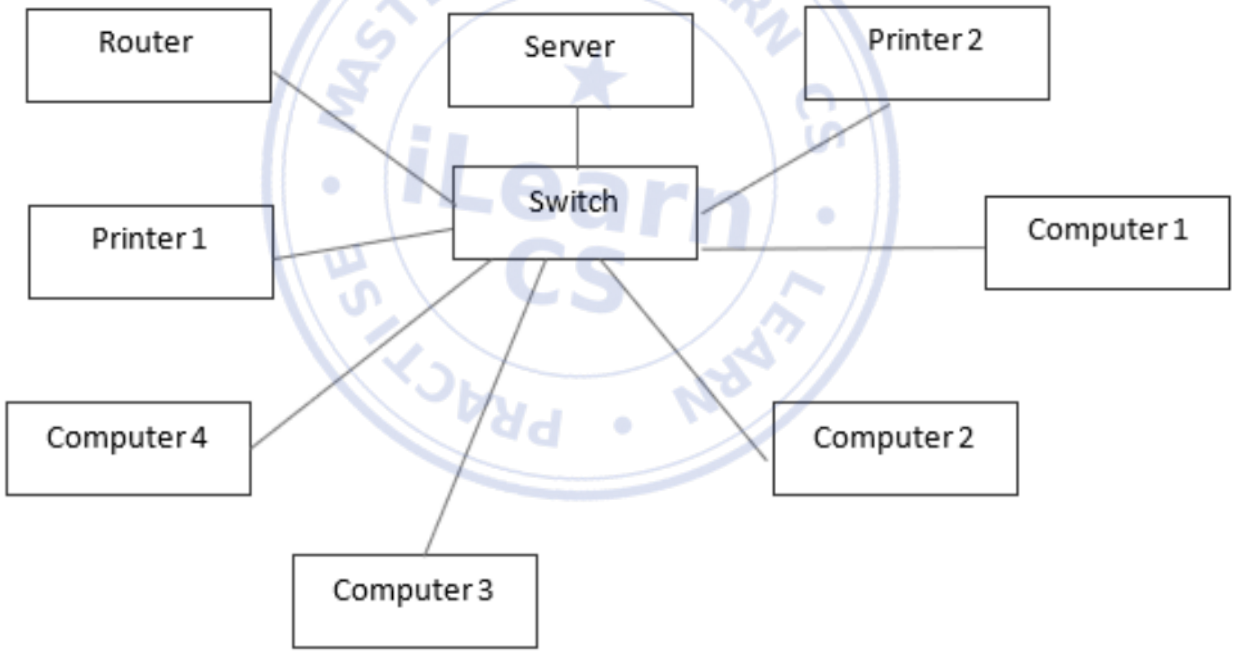
• Encodes/scrambles data
• ... so if it is intercepted it cannot be understood
• Algorithm/key is required to decode the data</td><td>2</td></tr>
</table>

Total: 75 marks

Question 5 · May/June 2025 Paper 12, original Q6 (a),(b),(c),(d),(e) [19 marks]

Question	Answer	Marks
5(a)	**1 mark** each to **max 2** • In a WAN devices can connect over a larger geographical area in a LAN devices connect over a smaller geographical area • A WAN may use non-dedicated hardware the hardware in a LAN is usually dedicated • A WAN uses external / not company owned transmission media // the transmission media for a LAN is usually company owned	**2**
5(b)	**1 mark** each to **max 4** e.g. • Land is split into cells designed for maximum line of sight • Each cell has a tower with an antenna which receives and transmits data • Data is transmitted between the tower and the phone • Data transmission is wireless using low power radio signals/frequencies • Multiple devices can communicate simultaneously with the same tower	**4**

5(c)	**1 mark** for each correct word • **full stops** • **hexadecimal** • **zeros** • **128** • **dynamic** • **private** IPv4 is made of four groups of 8-bit numbers separated by **full stops**. Each IPv4 address is 32 bits. IPv6 is made of eight groups of 4 **hexadecimal** numbers separated by colons. Multiple consecutive groups of **zeros** can be replaced with a double colon. Each IPv6 address is **128** bits. A **dynamic** IP address can change each time the computer connects to a network. A **private** IP address can only be accessed by other devices in the same LAN and is assigned by the router within the LAN.	**6**

5(d)	**1 mark** each • Server directly connected to the switch • Four labelled computers all connected directly to the switch • Two printers each directly connected to the switch / server / a computer • Router / modem directly connected to the switch / server for internet access e.g. Example only  	**4**

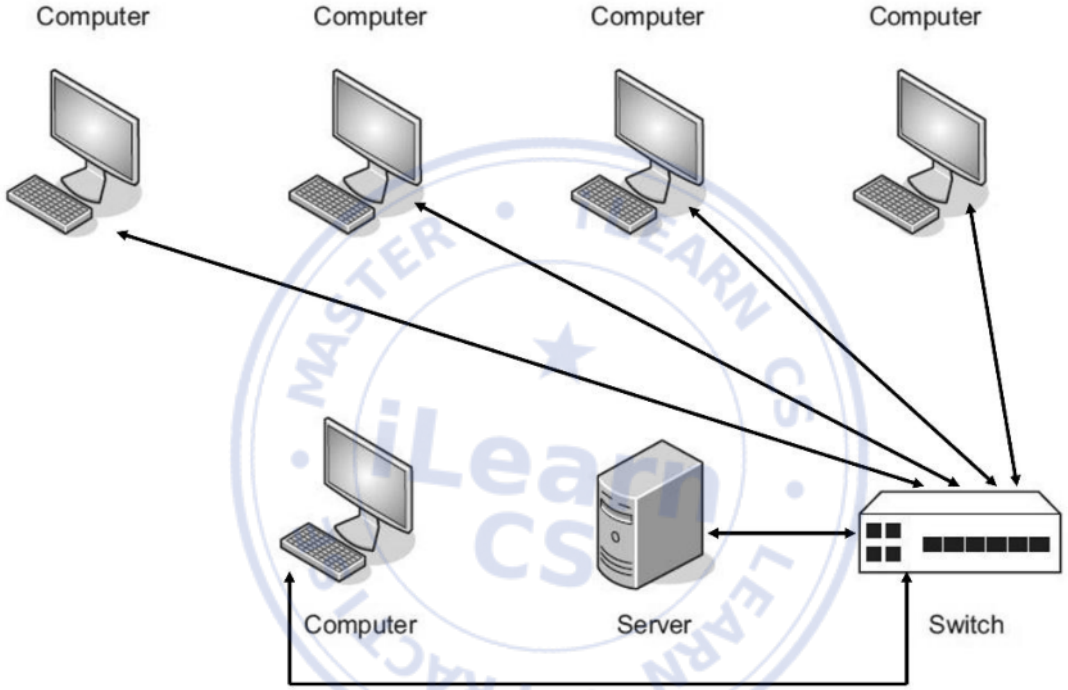
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5(e)	**1 mark** each to **max 3** e.g. • To store the MAC addresses of devices connected to it • To receive packets from devices • To forward packets directly to the intended recipient • To provide a central point of connection for the network • to enable connected devices to communicate	**3**

Question 6 · Oct/Nov 2025 Paper 12, original Q5 (a),(b),(c),(d),(e) [13 marks]

Question	Answer	Marks
6(a)	**1 mark** per bullet point, **max 2** marks • Covers a small geographical area • Normally privately owned / dedicated infrastructure • Can use wired or wireless communication	**2**

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6(b)	**1 mark** per bullet point, **max 2** marks • All computers directly connected to switch and no other connections • Switch connected directly to server  	**2**
6(c)	**1 mark** per bullet point, **max 3** marks • The sending device sends the packets to the switch • The switch checks the destination address on each packet • … and sends them **only** to the intended receiver	**3**

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6(d)	**1 mark** per bullet point, **max 3** marks • The use of CSMA/CD // Carrier Sense Multiple Access with Collision Detection • The workstations listen to the communication channel • ... and send data only when there is no data being transmitted / the line is idle • If a collision is detected, transmission is aborted • …and a jamming signal is sent • The workstation calculates a random wait time before trying to re-transmit • The random time is increased if there are multiple collisions	**3**
6(e)	**1 mark** for correct threat **1 mark** corresponding description **1 mark** for correct method of prevention	**3**

Threat	Description	Prevention
Malware / Virus	Malicious code that can alter/delete files	Anti-virus // Anti-malware // Firewall
Spyware	Records keystrokes which are sent to third party	Anti-spyware // Firewall // Anti-malware
Hacking	Gaining unauthorised access to a computer network / device	Authentication // Firewall

Phishing	Emails supposedly from reputable companies are sent to trick people into revealing personal information	Use a spam filter // Do not open emails from unknown sources
Pharming	Users are directed to a bogus website that looks legitimate to obtain personal information	Use a VPN // Anti-malware// Do not open links or download attachments